Day -4 : Subnetting Practical

1. Subnet Breakdown :

To split **192.168.10.0/24** into 4 equal subnets, borrow 2 bits ($2^2 = 4$). The prefix changes from /24 to /26, giving a subnet mask of **255.255.255.192**. Each subnet contains 64 total addresses (62 usable for devices).

Subnet #	**Network Address**	**Subnet Mask**	**Usable IP Range**	**Broadcast Address**
Subnet 1	192.168.10.0	255.255.255.192	192.168.10.1 – 192.168.10.62	192.168.10.63
Subnet 2	192.168.10.64	255.255.255.192	192.168.10.65 – 192.168.10.126	192.168.10.127
Subnet 3	192.168.10.128	255.255.255.192	192.168.10.129 – 192.168.10.190	192.168.10.191
Subnet 4	192.168.10.192	255.255.255.192	192.168.10.193 – 192.168.10.254	192.168.10.255

2. Device IP Allocation (Using Subnet 1) :

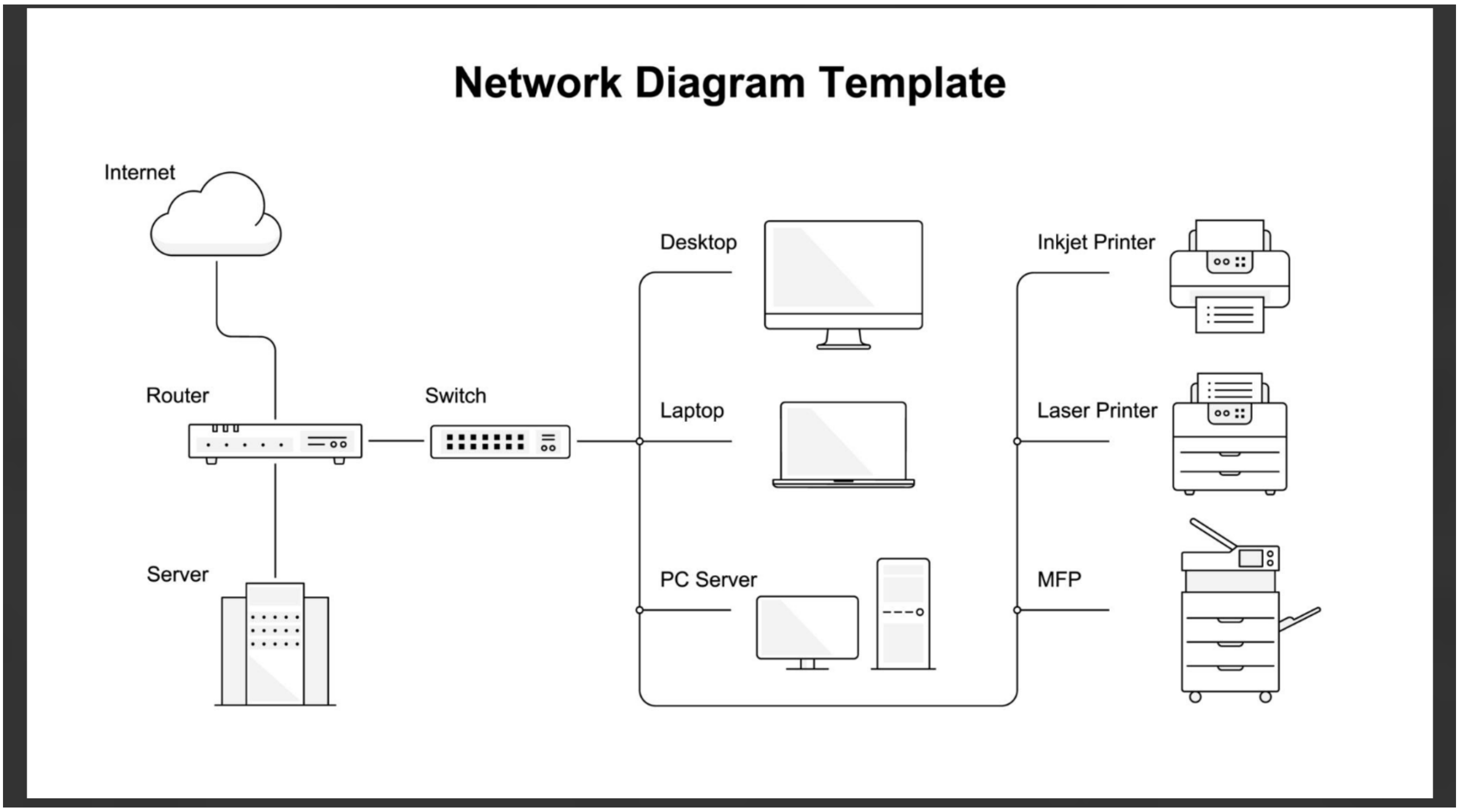
Using **Subnet 1** (192.168.10.0/26), assign valid IPs from the usable pool (.1 to .62):

- **Router (Default Gateway):** 192.168.10.1
- **Laptop 1:** 192.168.10.2
- **Laptop 2:** 192.168.10.3
- **Printer:** 192.168.10.4
- **Mobile Smartphone:** 192.168.10.5

(The network address .0 and broadcast address .63 are excluded).

3. Network Topology Diagram :

Below is a typical network topology layout illustrating how end devices connect to a central switch and router:



4. Step-by-Step Calculation Guide :

Imagine an IP address is like a street address: the **Network ID** is the street name, and the **Host ID** is the house number.

- **Step 1: Figure out how many bits to borrow**
  - Starting with /24, the first 24 bits belong to the network and 8 bits are left for host devices ($2^8 = 256$ addresses).
  - To make **4 equal subnets**, calculate how many power-of-2 bits are needed: $2^n = 4 \rightarrow n = 2$. Borrow **2 bits** from the host portion.
- **Step 2: Calculate the new Subnet Mask**
  - Add those 2 borrowed bits to the mask: $24 + 2 = \mathbf{/26}$.
  - In binary, the last octet was 00000000. Now it becomes 11000000.
  - Convert 11000000 to decimal: $128 + 64 = \mathbf{192}$.
  - **New Subnet Mask:** 255.255.255.192.
- **Step 3: Find the block size (magic number)**
  - Subtract the last octet mask value from 256: $256 - 192 = \mathbf{64}$.
  - This means subnets count up by **64** starting from 0: 0, 64, 128, 192.
- **Step 4: Calculate hosts per subnet**
  - With 6 bits left for host devices ($32 - 26 = 6$), each subnet has $2^6 = 64$ total slots.
  - Subtract **2 reserved addresses** (1 for the network name, 1 for broadcast): $64 - 2 = \mathbf{62}$ usable hosts.
- **Step 5: Fill in the boundaries**

- **Subnet 1:** Starts at .0. Next subnet starts at .64, so Subnet 1 ends with broadcast .63. Usable IPs: .1 through .62.
- Repeat this pattern for all 4 subnets!